$$m = \frac{Y_2 - Y_1}{X_2 - X_1}, \frac{Change\ in\ Y}{Change\ in\ X}.$$

Where (X_1, Y_1) and (X_2, Y_2) are two arbitrary points

Slope or Gradient of the line represents the ratio of the changes in vertical and horizontal lines.

The formula for constructing a straight line is

$$\boxed{(Y - Y_1) = m(X - X_1)}$$

If the two points are (0, 0) and (X, Y) then the formula is Y = mX

Example 12.1

Find the equation of a straight line which passes through two points (2, 2) and (4, –8) which are (X_1, Y_1) and (X_2, Y_2) respectively.

Note: For drawing a straight line, at least two points are required. Many straight lines can pass through a single point.

Solution

Here $X_1 = 2, \quad Y_1 = 2$

$X_2 = 4, \quad Y_2 = -8$

Formula for construction of straight line

$$\frac{Y - Y_1}{Y_2 - Y_1} = \frac{X - X_1}{X_2 - X_1}$$

Applying the values

$$\frac{Y-2}{-8-2} = \frac{X-2}{4-2}$$

$$\frac{Y-2}{-10} = \frac{X-2}{2}$$

$$2(Y-2) = -10(x-2)$$

$$2Y-4 = -10x+20$$

$$2Y = -10x+24$$

$$\boxed{Y = -5x+12}$$

–5 is slope, denoted by **m**

12 is **Y** intercept, or constant denoted by **c**

This is of the form Y = mX + c

Y = 12–5X when X = 0; Y = 12

When Y = 0; X = 12/5 = 2.4

(This line looks likes demand line in micro economics)

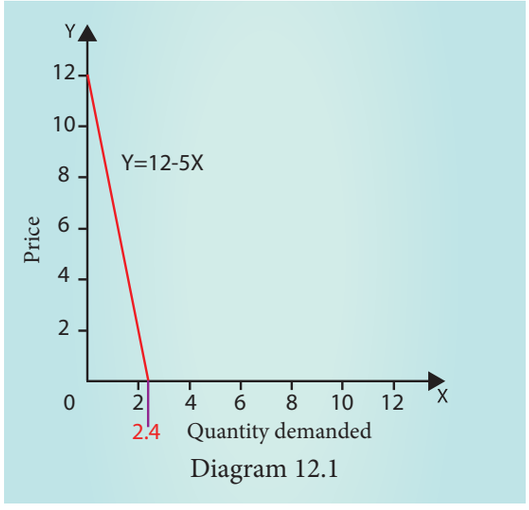


Diagram 12.1

12.2.3 Application in Economics

By applying the above method, the demand and supply functions are obtained.

Demand Function: $Q_d = f(P_x)$ where Q_d stands for Quantity demand of a commodity and P_x is the price of that commodity.

Supply Function: $Q_s = f(P_x)$ where 'Q_s' stands for Quantity supplied of a commodity and P_x is the price of that commodity.

In the example 12.1 the equation Y = –5X + 12 has been obtained. It is a linear function. Since slope is negative here, this function could be a demand function.

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